El AI: Voice AI in High-Pressure Aviation

The high-pressure aviation environment is ideal for **real-time, two-way Voice AI** driving immediate physical action — from passenger assistance to preventing costly flight delays.

Inbound Use Case

Terminal Personal Assistant — passengers call during transit for gate location, flight status, or emergency support.

Outbound Use Case

Urgency Agent for late passengers — identifies checked-in passengers not on board, calls to locate, directs to gate, and updates data source.

Outbound Agent: ROI & Strategic Rationale

Financial Breakdown

10-30x

ROI

Per month deployed

~$300

Agent Cost

Per month

$9K

Max Savings

2–3 delays prevented

Design cost: $400–$1,500 one-time. Monthly savings: ~$3,000–$9,000.

Why Voice AI over SMS/Push?

- Immediate urgency
- Two-way dialogue
- Real-time decisions

Why Outbound over Inbound?

- Direct and measurable ROI
- Operationally proven
- Improves on-time performance
- Reduces baggage costs & frees ground crew

Candidate Evaluation — STT · LLM · TTS

1

STT

Speech Recognition Accurate Hebrew transcription in a noisy airport environment is the hardest constraint.

- **Deepgram Nova-3** — ~100ms streaming latency, production-ready Hebrew, telephony-tuned noise handling
- **Whisper v3** — Excellent Hebrew accuracy, stronger under extreme noise, batch-only (~300ms)

2

LLM

Language Model Hebrew edge-case understanding and reliable function calling determine operational correctness.

- **GPT-4.1** — Excellent Hebrew incl. code-switching, improved instruction following and function calling, cheaper and faster than GPT-4o
- **GPT-4o** — Same Hebrew quality, proven production reliability for over a year, battle-tested function calling

3

TTS

Voice Synthesis Hebrew prosody quality is immediately audible — an unnatural voice breaks passenger trust in the first sentence.

- **ElevenLabs v3** — Best-in-class Hebrew prosody and intonation, real-time streaming, full emotional range
- **Azure TTS he-IL-HilaNeural** — Natural enough for production, reliable infrastructure, significantly cheaper

Component Selection & Architecture Strategy

Every component was chosen to minimize latency, maximize reliability, and handle Hebrew under real-world noise conditions.

1

STT — Deepgram Nova-3

Low latency streaming recognition (~100ms). Chosen over Whisper v3 to prevent cumulative delays.

2

LLM — GPT-4.1

Superior reliability in function calling over GPT-4o. Handles operational edge cases effectively.

3

TTS — ElevenLabs v3

Best-in-class Hebrew prosody over Azure he-IL. Delivers appropriate emotional range — calm to urgent.

Architecture Decision: Pipeline Architecture (over real-time models) — ensures observability, structured function calling, and proven Hebrew transcription under noise.

System Architecture & Call Flow

Cognitive Agent Flow (4 Steps)

1 **Fetch**

2 **Pickup**

3 **Locate**

4 **Check Edge Cases**

5 **Companions**

Check if companions are together

6 **Close**

Summarize and hang up

Agent Toolkit

- `get_passenger_data` — fetches passenger record before first word
- `summarize_call` — writes outcome + summary back to sheet
- `transfer_to_human` — live call transfer
- `end_phone_call` — explicit hangup (no-answer only)

Shortcut Rules

- Hebrew only
- Strict one-sentence limit per turn
- Gender-neutral plural forms only
- *Gate letters spelled in Hebrew (B07 → "גייט בי שבע")*

Live System Verification

All call summaries are automatically written back to Google Sheets using the `summarize_call` tool.

Passenger	Outcome	Summary (סיכום)
Dana Levi (LY315, B14)	✅ On Route	הנוסעת בדרכה לגייט
Moshe Avraham (LY315, B14)	❌ Canceled (Baggage Offloaded)	הנוסע לא טס והכבודה תורד מהמטוס
Rina Goldberg (LY447, A03)	🔄 Transferred to Agent	הנוסעת הועברה לנציג לסיוע רפואי/נגישות

The system successfully handled all three passenger scenarios — routing, cancellation with baggage offload, and medical/accessibility escalation — in real time.